

GEO Gravity Dynamics Lab

Experimental Orbital Dynamics and Phase-Modulated Effective Gravity

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Abstract

This work presents an experimental orbital dynamics study derived from the GEO framework (Hidden Geometry). The laboratory explores whether phase-modulated geometric operators can generate coherent orbital structures that differ from classical Newtonian closed trajectories.

Introduction

This repository presents an independent and reproducible orbital dynamics study based on effective geometric operators derived from the GEO framework. The primary objective is not to replace Newtonian gravity or General Relativity, but to investigate whether simple phase-dependent geometric modulations can generate coherent orbital structures with cumulative precession and non-closed trajectories.

Experimental Law

$F_{\text{GEO}}(r, \theta) = (G \cdot M_{\text{eff}}(r) / r^2) \cdot \cos(2\theta(r))$ The phase operator $\cos(2\theta)$ introduces effective dynamic geometric modulation.

Main Experimental Results

The study consistently reproduces: • rosette-type orbital structures, • cumulative orbital precession, • radial drift, • phase-modulated force evolution, • synchronized geometric operators, • and coherent curvature oscillations.

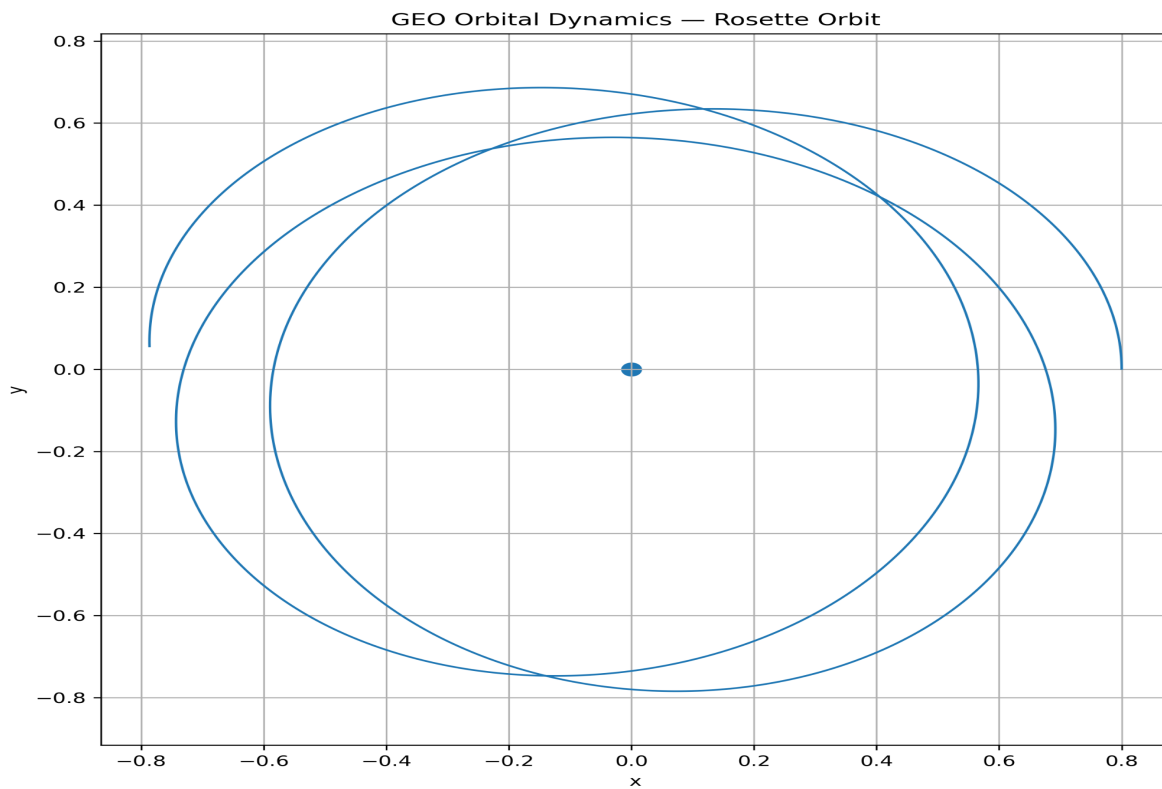


Figure 1 — GEO Rosette Orbit

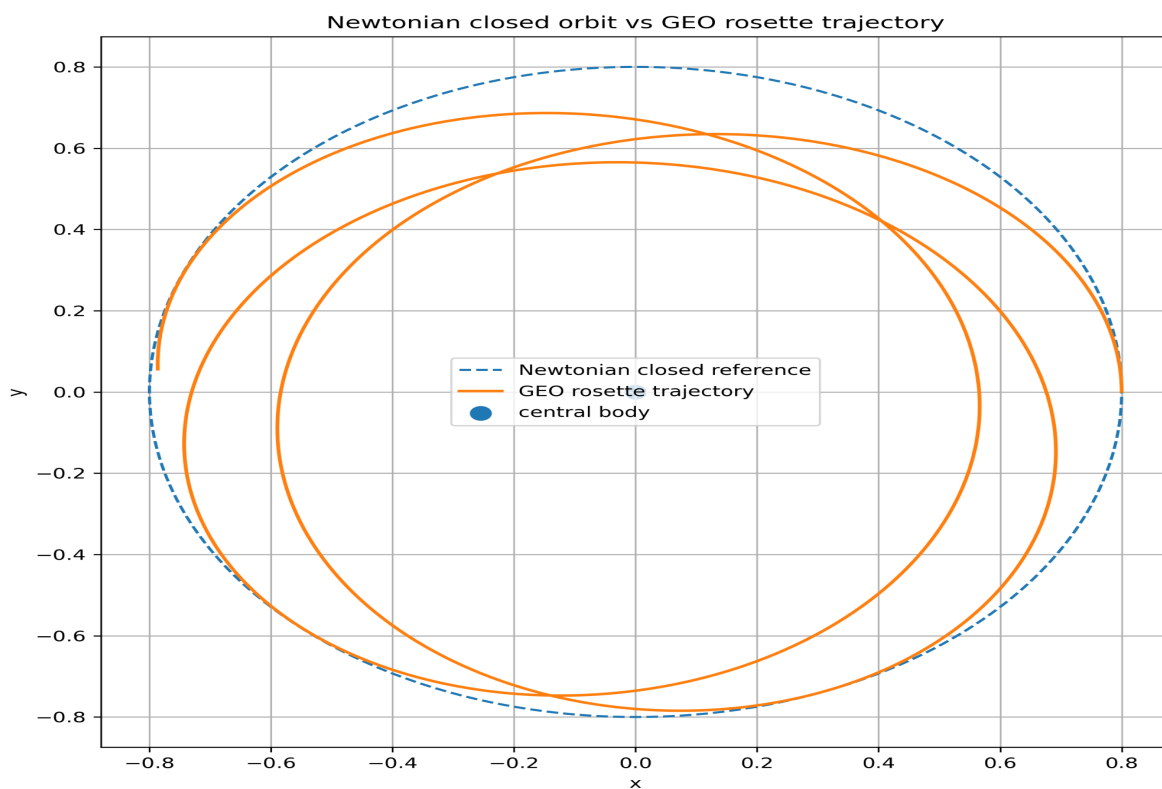


Figure 2 — Newtonian Closed Orbit vs GEO Rosette

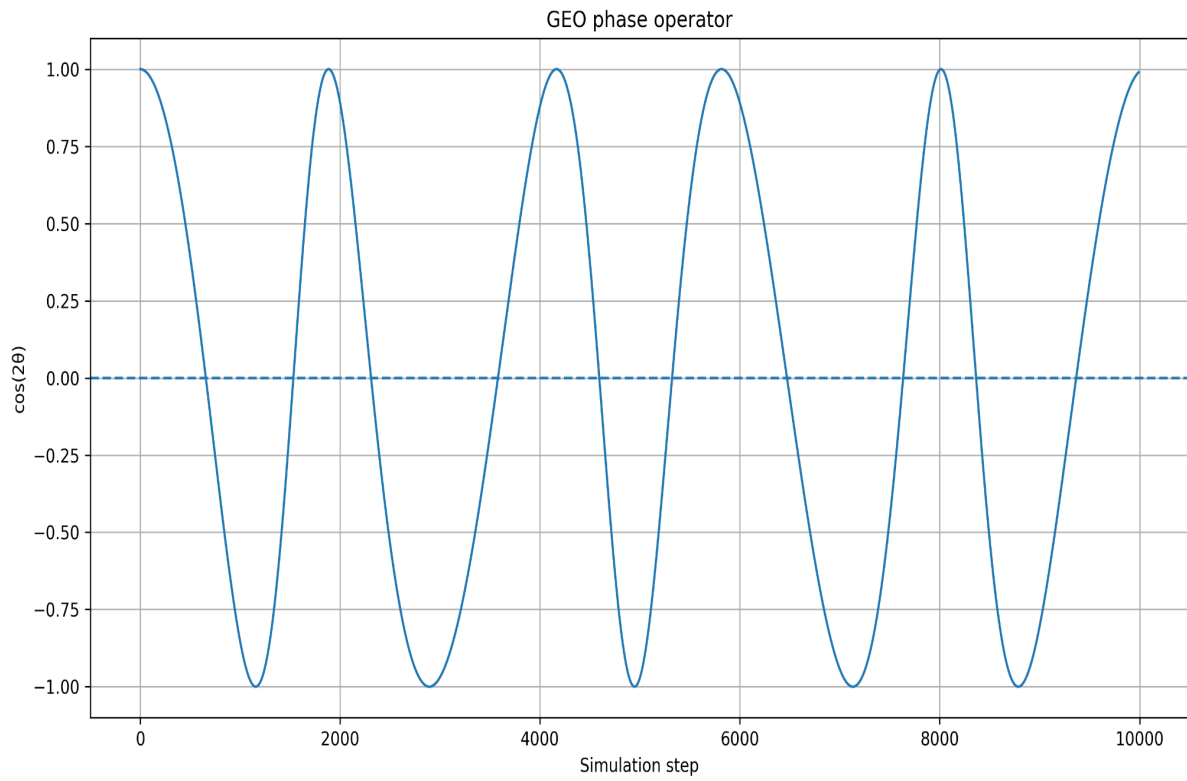


Figure 3 — GEO Phase Operator

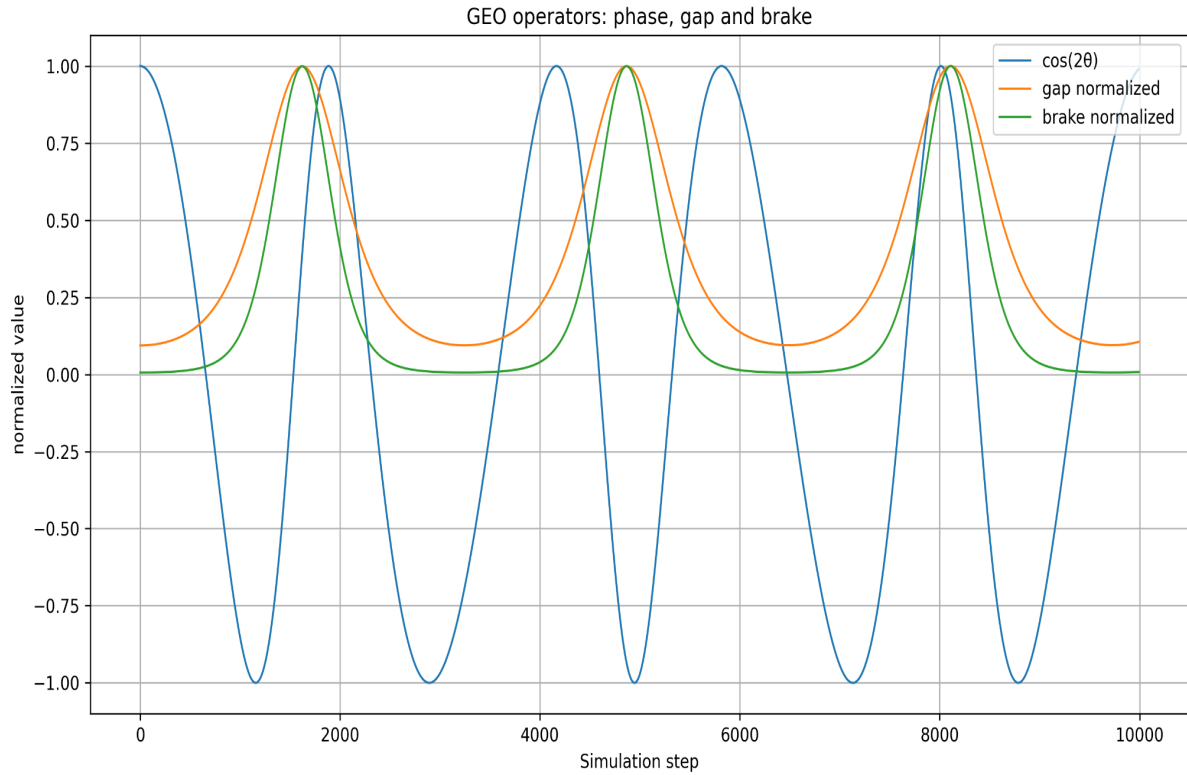


Figure 4 — GEO Phase, Gap and Brake Operators

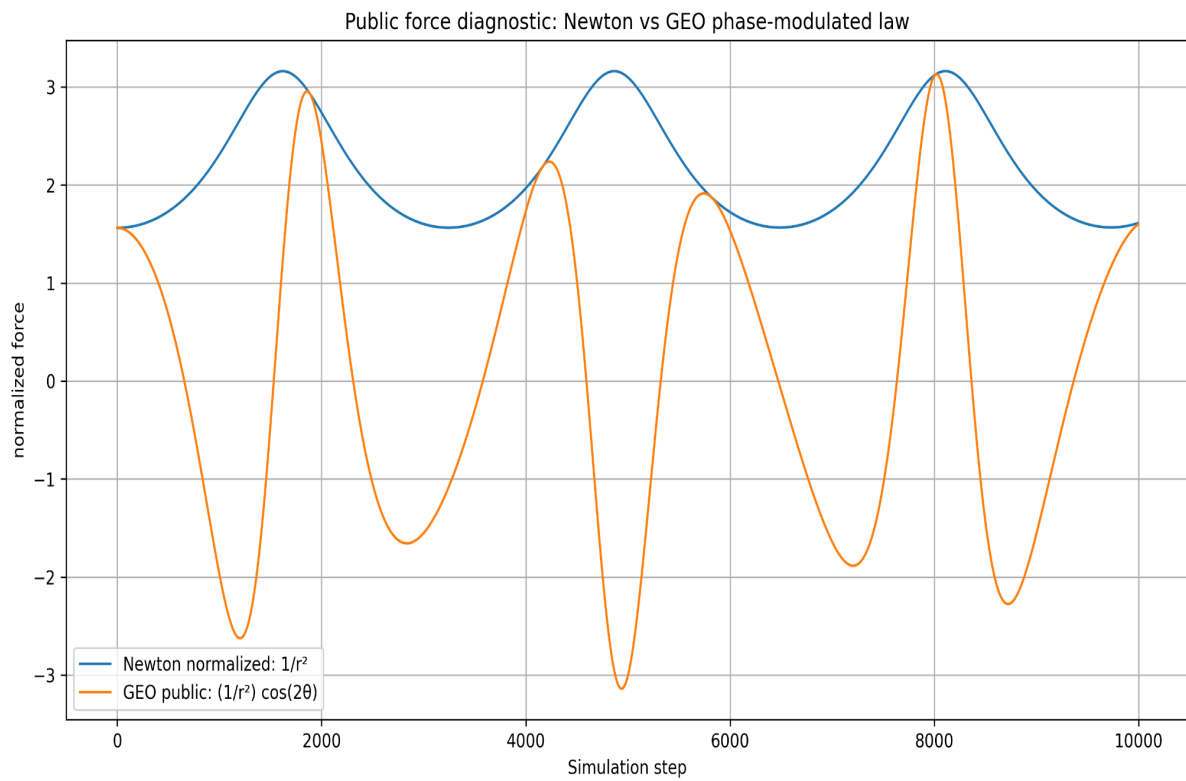


Figure 5 — Newton vs GEO Effective Force

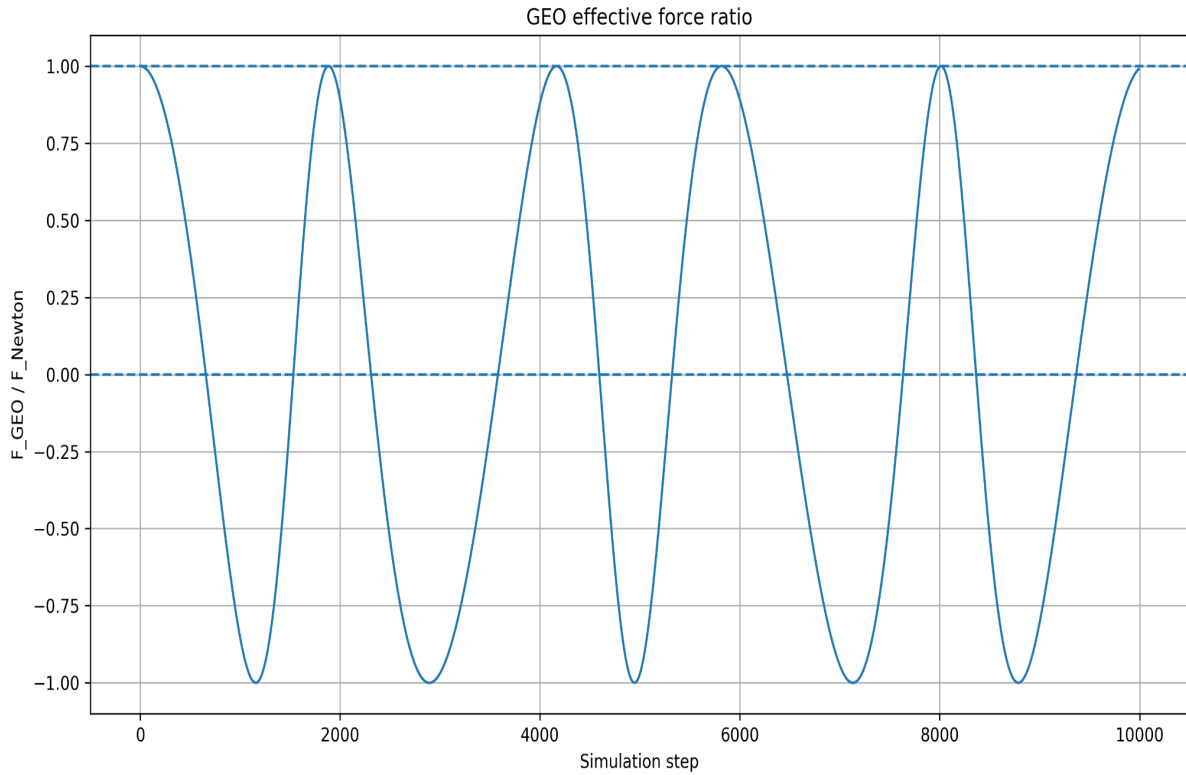


Figure 6 — GEO Effective Force Ratio

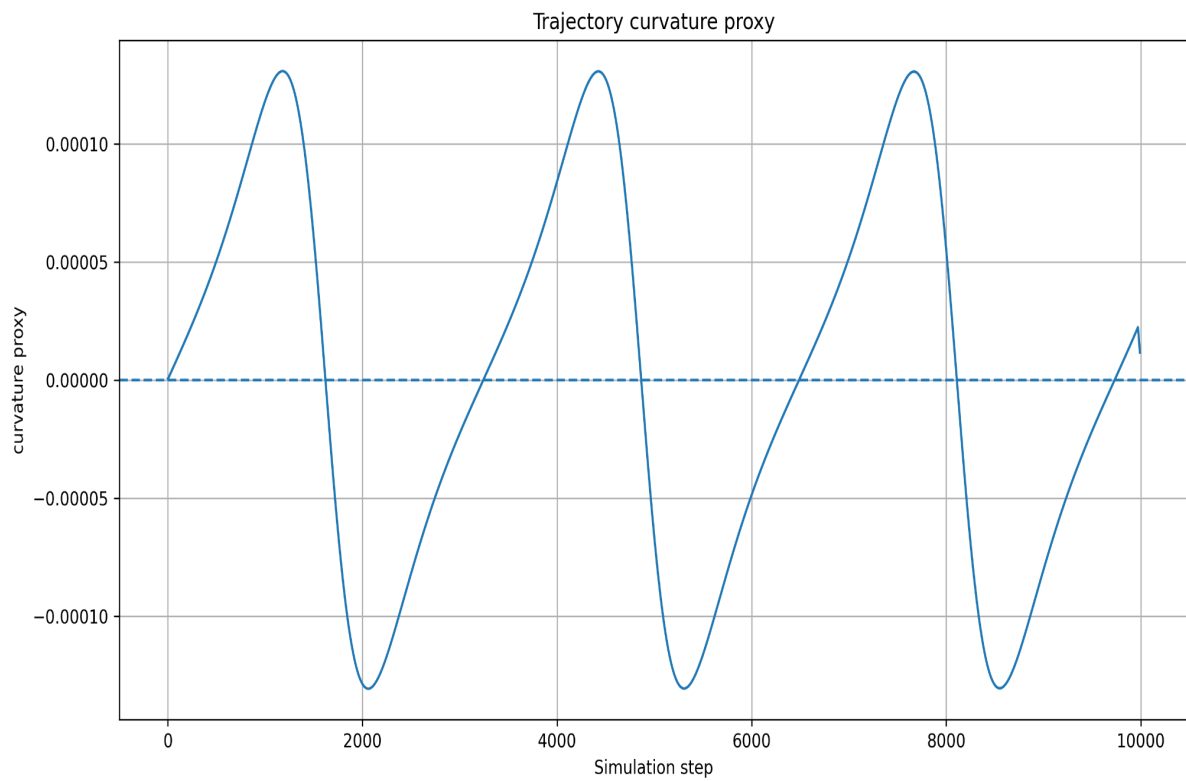


Figure 7 — Curvature Proxy

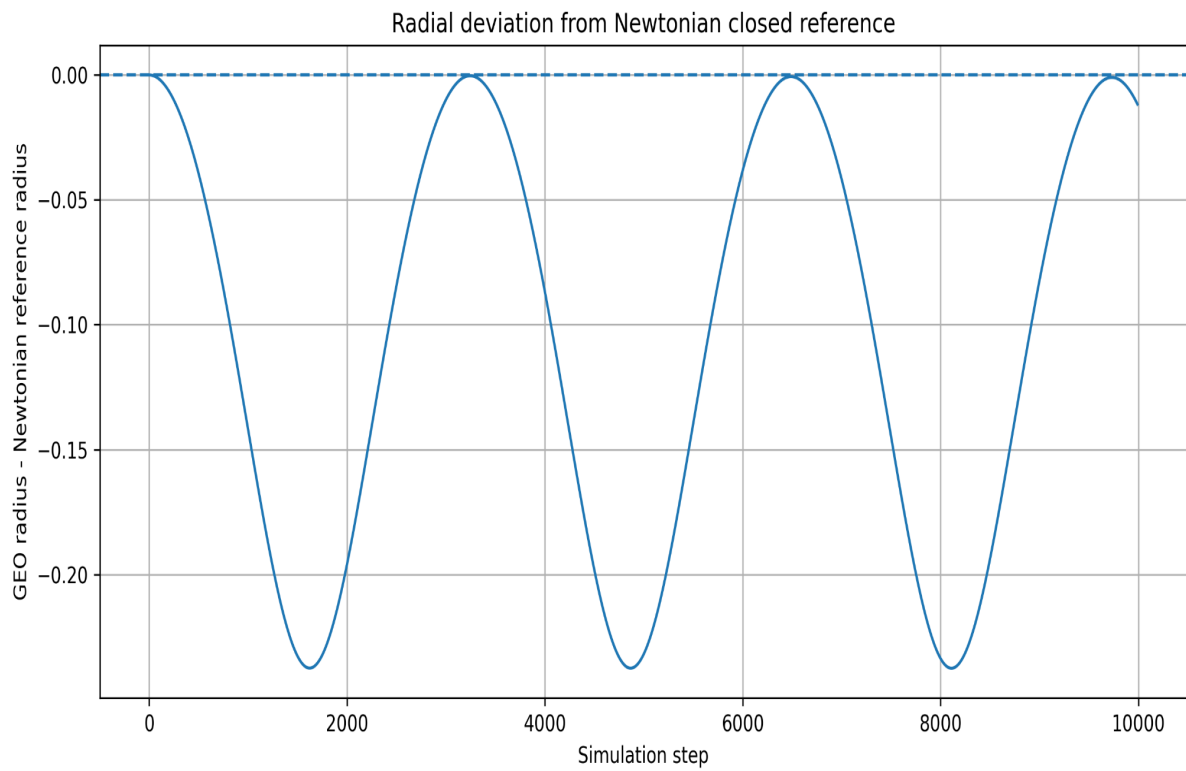


Figure 8 — Radial Drift

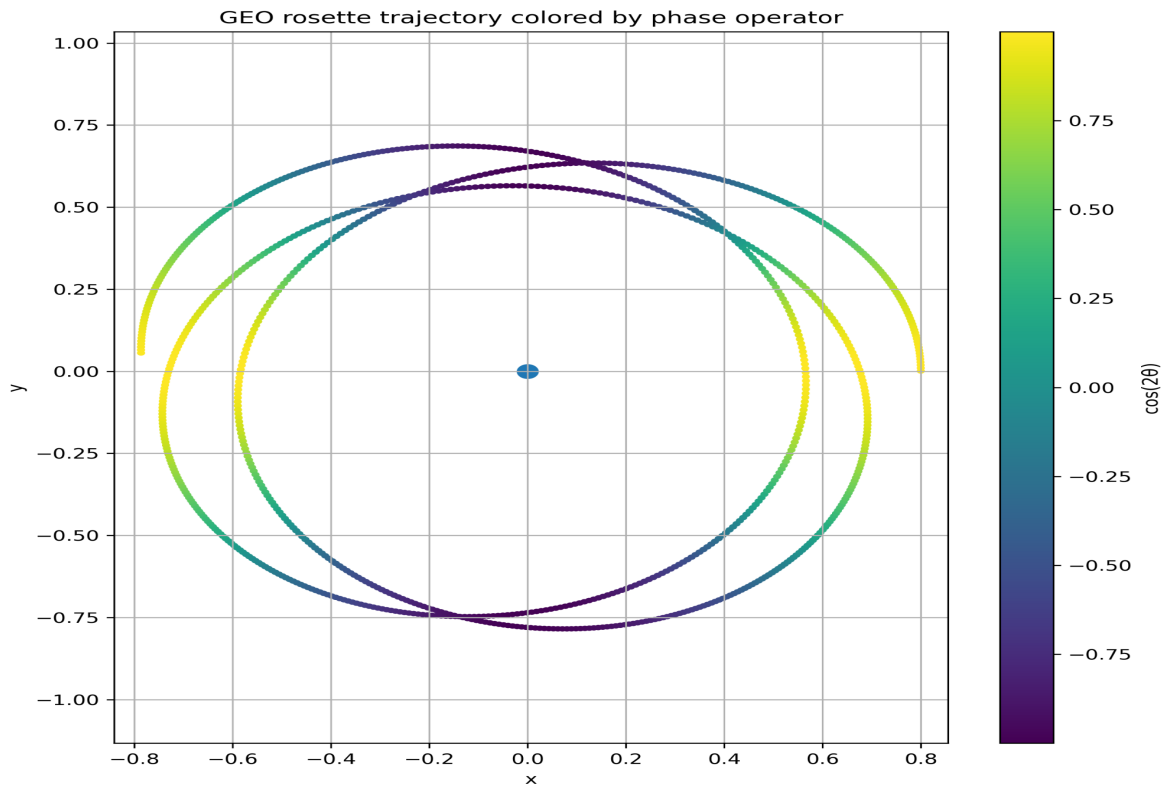


Figure 9 — GEO Rosette Colored by Phase

Discussion

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The resulting trajectories exhibit coherent non-closed orbital evolution, cumulative precession, radial drift, and synchronized phase-dependent modulation. Gap and brake operators remain correlated with transition regions and curvature changes, suggesting that the observed behavior preserves internal geometric structure.

Conclusion

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This experimental laboratory documents reproducible orbital structures generated by phase-modulated effective gravity operators derived from the GEO framework. The repository is intended as an open reproducible laboratory for independent mathematical inspection and discussion.

References

References

- GEO Hidden Geometry Framework — <https://github.com/LeoTorreblanca/GEO-hidden-geometry-framework>